Digital Ethics in Data Science

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Understanding the Responsibility Behind the Data

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Introduction to Digital Ethics

Moral Principles

Guiding the use of digital technology.

Key Focus Areas

Data privacy, fairness, transparency, and accountability.

Importance in Data Science

Data scientists handle sensitive and impactful information.

Ethical Challenges in Data Science

1. **Privacy Issues**
   Privacy Violations
2. **Inequality Concerns**
   Bias and Discrimination
3. **Transparency Questions**
   Lack of Transparency
4. **Data Consent**
   Unconsented Data Collection
5. **Analytics Misuse**
   Misuse of Predictive Analytics

Data Privacy and Consent

Informed Consent

Data should be collected with informed consent.

Minimal Collection

Minimize data collection to what's necessary.

Data Security

Secure data storage and access control.

Regulations

Examples: GDPR, Cookie Consent Banners

Bias and Fairness

1. **Algorithmic Bias**

   When a model unfairly favors one group.

2. **Sources**

   Biased data, poor feature selection, lack of diversity.

3. **Solutions**

   Fairness audits, inclusive datasets, bias detection tools.

Transparency and Explainability

User Rights

Users have the right to understand AI decisions.

Trust

Builds trust and accountability.

Interpretable Models

Use interpretable models and provide explanations (XAI).

Accountability in Data Science

1. Responsibility lies with data scientists, engineers, and organizations.

2. Best Practices: Ethical review boards, documentation, audits.

Real-World Examples

Recidivism Algorithm

COMPAS Recidivism Algorithm – racial bias.

Facebook Data Misuse

Cambridge Analytica – misuse of Facebook data.

AI Hiring Bias

Amazon AI Hiring Tool – bias against women.

Ethical Guidelines and Frameworks

1. IEEE Guidelines, OECD Principles, Data Ethics Framework (UK).
2. Core Principles: Beneficence, Non-maleficence, Autonomy, Justice.

Best Practices for Ethical Data Science

1

Privacy and Fairness

Design for privacy and fairness.

2

Model Auditing

Regularly audit models.

3

Diverse Engagement

Engage diverse stakeholders.

4

Transparency

Ensure transparency at all stages.

5

Fairness Assessment

Ask: Is this fair and necessary?

Conclusion

1. **Responsibility**

   Data science comes with great responsibility.

2. **Ethics**

   Ethical practices ensure fair and just technology.

3. **Decision-Making**

   Every data scientist is an ethical decision-maker.

References

GDPR

– https://gdpr.eu/

IEEE Design

Ethically Aligned Design

Harvard Center

Berkman Klein Center for Internet & Society

Thank You